SDC HACKATHON 2026

Phase 2: Final Project & Prototype Report

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 1 November 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Final.pdf" together with your pitch deck and a working deployment link. Answers must comply with the word limits stated for each open-ended question. Section 3 requires concrete, verifiable evidence. Unsupported claims will not be credited by judges.

1. Team & Project Overview

Team Name	Qart.ai
Project / Product Name	Qart.ai: AI-era reskilling for Malaysia's GBS finance workforce
Final Live Deployment URL	https://qartai.replit.app
Codebase Repository URL	https://github.com/Moo7x/qart.ai
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2. Project Description

[Maximum 250 words]

Qart.ai takes a person working in, or entering, Malaysia's Global Business Services (GBS) finance and accounting sector from "who I am today" to "where I can be in the AI era". A person uploads a CV; personal details are removed and gpt-4.1-mini reads their role and skills, quoting the CV sentence behind each claim. A daily pipeline collects dated public evidence from seven sources (OpenDOSM, DOSM job ads, Anthropic Economic Index, ILO GenAI exposure, O*NET, TalentCorp's GBS study, MyCOL). From it, Qart makes its own forecast: a trained model predicts AI use on the 320 tasks of 17 occupations, backtested models forecast each job group's demand, and 27 GBS finance roles are labelled Growing, Transforming, Shrinking, Emerging or Stable, with the evidence and its strength. Roles are ranked by an explainable fit (the person's skills against each role's sourced requirements) and the forecast. The reskilling path is built from the skills the person already has: each one is measured by how much AI already does on its tasks and becomes "check and steer AI-run work" or "do it faster with AI", with checked courses and videos on doing that work with AI. Choosing a role replaces it with the path to that role. Every role, skill and step is tagged to SDG 4.4, 4.3, 8.2 or 8.5 (12.6 for green skills) with a reason; two scenarios (moderate, fast AI) change weights, forecast and ranking. Every number is labelled observed, assumption or estimate; people make every decision.

3. Upgraded Product & Technical Execution

3.1 Final System Architecture

Provide a detailed description (and, if possible, an attached technical diagram) of your fully built system. Explain how your application integrates GenAI APIs, database structures, and backend/frontend frameworks into one unified solution.

Stack: FastAPI (Python 3.12) and SQLAlchemy Core over Postgres on Replit (SQLite locally); React + TypeScript + Vite + Tailwind front end served by the same app; a RunPod serverless GPU worker (BAAI/bge-m3 embeddings, bge-reranker-v2-m3); OpenAI gpt-4.1-mini for reading CVs, checking resources and the scout agent.

1. Evidence pipeline (src/pipeline, src/collectors): a scheduled Replit deployment runs every collector daily (GitHub Actions triggers it too); each observation is stored with source, date, URL and quoted span as an immutable version; a failed source keeps the last good snapshot.
2. Forecast layer (src/forecast): a task-level AI-use model (BGE-M3 features; ridge, random forest, gradient boosting, LightGBM and XGBoost compared by 5-fold cross-validation) predicts the automation-style share of AI use for unobserved tasks; DOSM job-group demand is forecast by the model that wins a rolling backtest against "no change"; roles get categories only when two of three AI evidence kinds agree.
3. Matching (src/match): fit = 100 × Σ w·min(c/r,1) ÷ Σ w over sourced requirements; scenario weights shift with observed AI use; roles are ordered by fit × outlook factor; "what would raise it" re-scores exactly.
4. Reskilling (src/pathway): one path, built from the person's own skills and their AI pressure, or to a chosen role; resources come from a catalogue job that retrieves, ranks and quote-checks courses and videos on doing each skill with AI.
5. Guardrails: schema-checked model answers, quotes verified in code, no model-written numbers, labelled fallbacks, PII scrubbing, a change test that edits evidence without storing it, and a proposal flow so new evidence never changes a person's results silently.

3.2 Implementation & Hackathon Deliverables

Describe what was successfully built, integrated, and deployed during the 48-hour physical sprint. Highlight any specific problem-solving steps your team took to pivot or improve the project from the Phase 1 preliminary design.

[Maximum 300 words]

Built and deployed in Phase 2: the full six-stage journey live at qartai.replit.app. Beyond the Phase 1 prototype we added (1) Qart's own forecast: a trained task-level AI-use model (gradient boosting, cross-validated error 11.2 points against 16.2 for the average) and backtested demand forecasts, turned into five plain role categories for 27 GBS finance roles; (2) matching that uses the forecast, with an external check: synthetic people built from each current role's typical skills find a TalentCorp-listed career move in their top 3 for 9 of 11 roles; (3) a reskilling path built from the person's own skills and where AI is taking them, replaced by the chosen role's path; (4) a catalogue job that finds courses and YouTube videos on doing each skill with AI, ranked by BGE-M3, a reranker and a learned ranker and kept only with quotes checked on the page or captions; (5) a change test that edits AI-use or job-market figures and shows which roles change category; (6) fairness, privacy (export, delete, 90-day retention), cost per person and a load test.

Pivots from Phase 1: the three-track plan (AI basics, without AI, with AI) became one path with AI inside every step, after user feedback; demand data moved from display into the ranking; and the same CV now always gets the same reading after we found the model could link one sentence to different skills on different runs.

4. Environmental & SDG Impact Verification

Validate the real-world impact of your built solution. Show concrete data that validate your environmental or social value proposition, compared against the baseline and KPI defined in previous phase submission.

*The data and evidence you provide in this section will be used to assess **SDG Impact, Technical Execution and Innovation scores.** Vague claims or missing verification will weaken your score across all three categories.*

Metric Category	**Your Response**
Target SDG Metric Achievement *Direct comparison between your Phase 1 baseline and your Phase 2 finalized prototype numbers* *e.g., "Achieved a 24.5% reduction in computational execution overhead against the 20% initial target."*	Coverage: 73% → 83% of required skills have a verified free course (35 of 42; target 90% not reached). New in Phase 2: 23 of the 42 also have checked material on doing the skill with AI. Projected readiness (estimate): synthetic profiles Hafiz, Mei and Nur move from an average best-role fit of 32.1 to 55.8 after their plans (+23.7; target +50 not reached: in Phase 2 study alone reaches level 2 and higher levels need practice at work). Time to

Metric Category	Your Response
	a plan (observed, app timer): live site 3 of 3 people under 5 minutes, median 109 s; 5 local test sessions 1 of 5, median 718 s; the 10-person test is not yet done. Traceability: every role, skill, course and forecast carries its source and date and an observed / assumption / estimate label.
Verification Evidence Method *How was this data verified?* *e.g., codebase algorithmic profiling logs, hardware power-meter simulation readings, or database transaction counters.*	Coverage: computed from the course catalogue (data/packs/gbs_fa). Readiness: the same engine on the fixed synthetic personas (data/samples/personas.yaml), regression-tested by tests/test_api_golden.py. Ranking quality: tests/test_ranking_check.py. Fairness: tests/test_fairness.py (profiles differing only in protected attributes get identical results). Usage and cost: the usage_event counter and scripts/cost.py (docs/COST.md). Load: Locust, docs/LOAD_TEST.md (50 people, 0 failures). Time to a plan: measured by the app's timer (CV upload to destination chosen), shown on the Status page.
Socio-Economic / Civic Value *How does your deployed application translate into a scalable, real-world solution capable of being supported by civic or corporate partners?*	Directly serves SDG 4.4 (job-relevant skills) and 8.5 (decent work): 32% of Malaysia's GBS workforce, about 89,000 people, are in roles TalentCorp rates highly impacted (TalentCorp 2024, p. 48). Qart gives each of them, free, an explained forecast of their job and a plan built from their own skills, using free courses and HRD Corp routes. Career centres, HR teams and public agencies (TalentCorp, HRD Corp, PERKESO MYFutureJobs) can adopt it to advise many people at once.
Deployment & Commercial Feasibility *e.g., modeled as an API service plugin designed for easy integration into existing corporate or institutional systems.*	Modelled as a web app plus API for career centres and HR portals: one pipeline serves every user; per-person cost about USD 0.002 (language model and GPU, measured usage × list prices); hosting on Replit Autoscale, GPU on RunPod serverless (scale to zero). New fields are added as data packs, not code.

4.1 Mathematical Proof / Code Logic Snippet (Optional, but holds bonus marks)

Paste the specific code section, mathematical formula, simulation log output, or calculation logic that proves the impact claims made above.

This proof is graded across three categories at once: ***Impact and Feasibility*** *(does the math support your claim),* ***Technical Execution*** *(does it reflect what you actually built), as well as* ***Innovation and Creativity*** *(is the approach original).] Do not leave this section generic or copied from Phase 1.*

Fit (src/match/engine.py): fit = 100 × Σ_k w_k·min(c_k / r_k, 1) ÷ Σ_k w_k
(c = the person's claimed level, r = the role's required level from cited sources, w = core 3 / upskill 2 / support 1, then the scenario weight below)
Scenario weight: w' = w × max(0.1, 1 + d × s × a) (d = +1 AI-complementing, −1 routine; s = 0.5 moderate / 1.0 fast; a = observed automation-style AI use on the skill's tasks)
Order: direction score = fit × outlook factor (growing 1.10, emerging 1.05, stable 1.00, transforming 1.00, shrinking 0.85)

Measured logs (python -m src.match.validation; tests/test_ranking_check.py):
fit only: 82% of 11 roles find a TalentCorp-listed move in their top 3, recall 0.53
fit × outlook: 73%, recall 0.50 (the outlook lifts growing roles; the test floor is 70%)
Task model version 2 (task_model_version table): 2,691 training tasks (Economic Index 2026-05, BGE-M3 features), 5-fold MAE: mean 16.15, ridge 11.37, random forest 11.98, gradient boosting 11.22 (chosen), LightGBM 11.34, XGBoost 11.26.
AI resources (docs/RESOURCE_EVAL.md): precision at 3 — source order 0.21, dense 0.28, reranker 0.31, learned ranker 0.34 (CV AUC 0.73); 41 checked resources for 28 skills.

5. Market Viability & Future Scalability Plan

SDC aims to encourage solutions that transition from innovations into market-ready applications. If given access to corporate funding, industry grants, or partnership opportunities (e.g., through SDC's Rotary network or joint research initiatives at XMUM), how would your team scale this prototype to process real-world data at volume?

[Maximum 250 words]

First users and channel: TVET and university career centres, and GBS employers' HR and L&D teams, which advise the people Qart serves (working adults at risk, graduates, career switchers). TalentCorp's MyMahir platform and HRD Corp are natural partners: Qart turns their studies and programmes into a personal, explained plan. Scale path: (1) a public API and embeddable widget so a career centre or HR portal can add the journey; (2) more field packs (the code is field-independent: a pack declares roles, skills, tasks and courses), starting with GBS IT and shared services; (3) Malaysian demand at role level by licensing job-posting data instead of broad DOSM groups; (4) a human review of the resource labels and a larger labelled set to retrain the rankers on RunPod. Running cost is low: about USD 0.002 per person for the language model and GPU, so 10,000 people a month cost about USD 21 plus fixed hosting; the pipeline costs cents a day. Through SDC's Rotary network and joint research at XMUM we would pilot with one career centre and one GBS employer, measure time-to-plan and course completion, and publish the results.

6. Final Submission Checklist

Tick all boxes that apply before submitting.

- ☑ The application is fully compiled, live, and accessible via the deployment link provided in Section 1.
- ☑ The GitHub (or equivalent) repository contains a clean, well-documented README with installation instructions.
- ☑ The GitHub (or equivalent) repository shows the history of incremental code updates made throughout the official live hacking window.
- ☑ The final pitch deck has been exported and saved as a flattened PDF file.
- ☑ All fields in this template have been completed and word limits respected.

7. IP & Originality Declaration

7.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No ☐ No change from Phase 1 declaration

If yes, briefly describe what was pre-existing and which team member(s) owned it:

7.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No ☐ No change from Phase 1 declaration

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
FastAPI, SQLAlchemy, React, Vite, Tailwind, scikit-learn, LightGBM, XGBoost	code	MIT / BSD / Apache-2.0
BAAI/bge-m3, BAAI/bge-reranker-v2-m3 (RunPod)	model	MIT / Apache-2.0
OpenAI gpt-4.1-mini (API)	model	OpenAI terms
O*NET 31.0, Anthropic Economic Index, ILO GenAI exposure index	dataset	CC BY 4.0
OpenDOSM, DOSM Job Market Insights, TalentCorp GBS study, MyCOL	dataset	open data / facts cited
YouTube Data API v3, Microsoft Learn Catalog API	API	provider terms

PyTorch, sentence-transformers, transformers (RunPod worker)	code	BSD-3-Clause / Apache-2.0
Full list with versions: SDC2026_Qart.ai_TechDeclaration.pdf, docs/DECLARATION_LOG.md	—	—

7.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No ☐ No change from Phase 1 declaration

If yes, please describe the nature of the agreement and which member it concerns:

8. Declaration

By submitting this report, the team leader confirms that the deployed application and data presented above are the original work of the team, completed within the official SDC 2026 hackathon period.

Kydyrgaliyev Meiirman (team leader)

1 November 2026